Rotational Symmetry

An object is said to have a *rotational symmetry* if it looks the same after being rotated about its centre through an angle less than 360°.

When an object rotates around a fixed axis if its appearance of size and shape does not change then the object is supposed to be rotationally symmetrical.

Rotational symmetry can be observed in the following *Fig. 4.4*

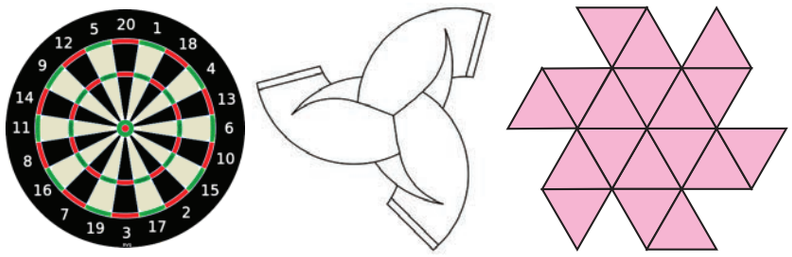


Fig. 4.4

The minimum angle of rotation of a figure to get exactly the same figure as original is called the angle of rotation.

The total number of times a figure coincides with itself in one complete rotation is called the order of rotational symmetry. We can only rotate the figure up to 360 degrees.

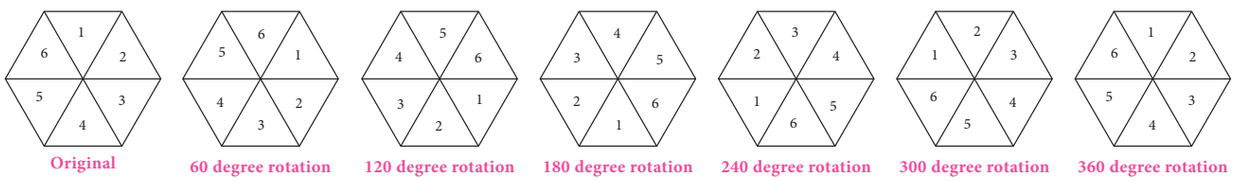
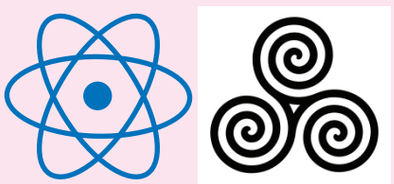


Fig. 4.5

It is important to understand that all figures have rotational symmetry of order 1, as can be rotated completely through 360° to come back to its original position. So we say that an object has rotational symmetry, only when the order of symmetry is more than 1. So, 2 is the smallest order of rotational symmetry.

Try these

1. Find the order of rotational symmetry of the following figures.



2. Find the order of rotational symmetry for an equilateral triangle.

Think

Can a parallelogram have a rotational symmetry?

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